

# RELATIVELY FREE JENNINGS QUOTIENTS AND TRUNCATED MODULAR GROUP ALGEBRAS

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**ABSTRACT.** Let  $p$  be a prime and let  $G$  be a finite  $p$ -group with Jennings series  $(M_n(G))_{n \geq 1}$ . We show that if  $G/M_k(G)$  is relatively free, that is, isomorphic to  $F/M_k(F)$  for a free group  $F$  of rank  $d(G)$ , then the isomorphism type of  $G/M_{p+k-1}(G)$  is determined by the truncated group algebra  $\mathbb{F}_p G/I^{p+k-1}$ , where  $I$  is the augmentation ideal. For  $k = 2$  the hypothesis is automatic, so  $\mathbb{F}_p G/I^{p+1}$  determines  $G/M_{p+1}(G)$  for every finite  $p$ -group  $G$ . In particular, the modular isomorphism problem has a positive answer for  $p$ -groups  $G$  with  $M_{p+1}(G) = 1$ . The proof works in the truncated free associative algebra. It combines the Lazard correspondence with a projection onto the Lie part defined through the restricted Poincaré–Birkhoff–Witt theorem. All logarithms and exponentials are applied only on the  $k$ -th power of the augmentation ideal, where they are  $p$ -integral.

## 1. INTRODUCTION

Let  $p$  be a prime, let  $G$  be a finite  $p$ -group, and let  $I = I_G$  be the augmentation ideal of the group algebra  $\mathbb{F}_p G$ . The modular isomorphism problem asks whether  $\mathbb{F}_p G \cong \mathbb{F}_p H$  implies  $G \cong H$  for finite  $p$ -groups  $G$  and  $H$ . The answer is negative for  $p = 2$  [GLMdR], and the problem is open for odd  $p$ ; see the survey [Mar].

A standard strategy is to show that  $\mathbb{F}_p G$  determines certain quotients of  $G$ . The *Jennings series* (also called the Brauer–Jennings–Zassenhaus series)

$$M_n(G) = G \cap (1 + I^n), \quad n \geq 1,$$

is adapted to this purpose. Since  $M_n(G) - 1 \subseteq I^n$ , the truncation  $\mathbb{F}_p G/I^n$  depends only on  $G/M_n(G)$ , and one may ask to what extent it determines  $G/M_n(G)$ . For groups with  $M_3(G) = 1$  the modular isomorphism problem has a positive answer by Passi and Sehgal [PS]. For odd  $p$  and groups with  $M_4(G) = 1$  it has a positive answer by Hertweck [He]. A systematic approach through the free group was developed by Röhl [Rö87, Rö90]. If  $G$  is  $d$ -generated, an automorphism of  $\mathbb{F}_p G$  lifts to an automorphism of the completed group algebra of the free group of rank  $d$ . Our argument works with a truncated version of this lifting.

For a group  $X$  we write  $d(X)$  for the minimal number of generators. For the free group  $F$  of rank  $d$  we put  $M_n(F) = F \cap (1 + I_F^n)$ , where  $I_F$  is the augmentation ideal of  $\mathbb{F}_p F$ . We say that a finite  $p$ -group  $G$  is *relatively free modulo  $M_k$*  if  $G/M_k(G) \cong F/M_k(F)$  with  $F$  free of rank  $d(G)$ .

**Theorem A.** *Let  $p$  be a prime and  $k \geq 2$ , and let  $G$  and  $H$  be finite  $p$ -groups. Suppose that  $G$  is relatively free modulo  $M_k$ , and that  $\varphi: \mathbb{F}_p G/I_G^{p+k-1} \rightarrow \mathbb{F}_p H/I_H^{p+k-1}$  is an isomorphism of algebras. Then*

$$G/M_{p+k-1}(G) \cong H/M_{p+k-1}(H).$$

*Moreover, the isomorphism can be chosen so that the map it induces  $G/\Phi(G) \rightarrow H/\Phi(H)$  corresponds, under  $g\Phi(G) \mapsto g - 1 + I_G^2$ , to the map  $I_G/I_G^2 \rightarrow I_H/I_H^2$  induced by  $\varphi$ .*

The hypothesis is visible in the algebra. Indeed  $d(G) = \dim I/I^2$ , and by Jennings' formula (1) the order of  $G/M_k(G)$  is determined by the dimensions of  $I^n/I^{n+1}$  for  $n < k$ . So  $H$  is relatively free modulo  $M_k$  as well; see Step 1 in Section 8.

That one further layer is determined, i.e. that  $G/M_{k+1}(G)$  is determined when  $G$  is relatively free modulo  $M_k$ , is a theorem of Röhl; see [HS, Theorem 5.8]. For  $p = 2$  this is exactly Theorem A. For odd  $p$ , Theorem A adds  $p - 2$  further layers.

For  $k = 2$  the hypothesis is automatic, because  $G/M_2(G) = G/\Phi(G)$  is elementary abelian of rank  $d(G)$ .

**Corollary B.** *Let  $G$  be a finite  $p$ -group. Then  $\mathbb{F}_p G/I^{p+1}$  determines  $G/M_{p+1}(G)$  up to isomorphism. In particular, if  $M_{p+1}(G) = 1$  and  $\mathbb{F}_p G \cong \mathbb{F}_p H$ , then  $G \cong H$ .*

*Proof.* The first statement is Theorem A with  $k = 2$ . For the second, note that  $|H| = \dim \mathbb{F}_p H = |G|$ . By the first statement  $G = G/M_{p+1}(G) \cong H/M_{p+1}(H)$ , so  $|H/M_{p+1}(H)| = |H|$  and therefore  $M_{p+1}(H) = 1$ .  $\square$

Since  $M_3(G) = 1$  and, for odd  $p$ ,  $M_4(G) = 1$  both imply  $M_{p+1}(G) = 1$ , Corollary B recovers, over  $\mathbb{F}_p$ , the theorems of Passi–Sehgal and Hertweck quoted above. If  $G$  has exponent  $p$ , then Jennings’ description  $M_n(G) = \prod_{ip^j \geq n} \gamma_i(G)^{p^j}$  reduces to  $M_n(G) = \gamma_n(G)$ . So Corollary B applies, for instance, to every group of exponent  $p$  and nilpotency class at most  $p$ .

**Method.** Write  $G$  and  $H$  as quotients of  $P = F/M_{N+1}(F)$  with  $N = p + k - 2$ . Then both truncated group algebras are quotients of the truncated free algebra  $T = A/\mathfrak{m}^{N+1}$ , where  $A$  is the free associative algebra over  $\mathbb{F}_p$  on  $X_1, \dots, X_d$  and  $\mathfrak{m}$  is its augmentation ideal. As in Röhl’s approach [Rö87, Rö90], an isomorphism between them lifts to an automorphism  $\theta$  of  $T$ . We must then replace  $\theta$  by an automorphism coming from the group  $P$ .

We argue by induction over the layers, which reduces the problem to unipotent automorphisms. For these we take the Lie part  $D_L$  of the logarithm  $D = \log u$  and exponentiate it to a group automorphism  $\sigma$ . The difficulty is that for  $N \geq p + 1$  neither  $\log u$  nor  $e^{D_L}$  is  $p$ -integral on all of  $T$ ; see Remark 9.2.

The key observation is that everything only needs to hold on  $\mathfrak{m}^k$ . The relation ideal lies in  $\mathfrak{m}^k$ , and the action of an algebra endomorphism on  $\mathfrak{m}^k$  only depends on the images of the generators modulo  $\mathfrak{m}^{N-k+2}$ . In the window  $N \leq p + k - 2$ , all the series involved have denominators prime to  $p$  on  $\mathfrak{m}^k$ . Section 9 explains where each consequence of  $N \leq p + k - 2$  is used, and what fails at  $N = p + k - 1$ .

**AI disclosure and authorship status.** I, the person whose name appears between question marks at the header of the document, do not claim to be the author of anything, neither the mathematical ideas nor the redaction, in this paper. It was all generated by Claude Opus 5.5 during one single day. I merely provided a reference on the topic ([He]) and asked for a generalization. There was previous attempt to prove Theorem A for  $k = 2$  by Ángel de Río and myself this summer with a free version of ChatGPT (aside of several AI-free attempts by myself), which ended up with a draft containing many of the ideas and arguments in this document up to Section 5 (included), but with enough mistakes to discourage us both at the time.

## 2. CONVENTIONS AND STANDARD FACTS

Throughout,  $p$  is a prime,  $k \geq 2$  and  $d \geq 1$  are integers, and  $F$  is the free group on  $x_1, \dots, x_d$ . Let  $\Lambda = \mathbb{Z}_{(p)}$ . We use the truncated logarithm and exponential

$$\ell(t) = \sum_{j=1}^{p-1} \frac{(-1)^{j+1}}{j} t^j, \quad \varepsilon(t) = \sum_{j=0}^{p-1} \frac{t^j}{j!}.$$

Let  $H(t_1, t_2) = \sum_{n \geq 1} H_n(t_1, t_2)$  be the Baker–Campbell–Hausdorff series, with  $H_n$  homogeneous of degree  $n$ , and put  $H_{<p} = \sum_{n < p} H_n$ .

### 2.1. Truncated identities.

**Lemma 2.1.** *Let  $z_1, z_2$  be elements of an associative  $\Lambda$ -algebra, for instance of an  $\mathbb{F}_p$ -algebra, such that every product of  $p$  factors from  $\{z_1, z_2\}$  vanishes. Then:*

- (a)  $\varepsilon(\ell(z_1)) = 1 + z_1$  and  $\ell(\varepsilon(z_1) - 1) = z_1$ ;
- (b) if  $z_1 z_2 = z_2 z_1$ , then  $\ell((1 + z_1)(1 + z_2) - 1) = \ell(z_1) + \ell(z_2)$ ;
- (c)  $\varepsilon(z_1)\varepsilon(z_2) = \varepsilon(H_{<p}(z_1, z_2))$ . In particular  $\varepsilon(z_1)\varepsilon(-z_1) = 1$ .

*Proof.* The corresponding identities in  $t_1, t_2$  hold in  $\mathbb{Q}\langle\langle t_1, t_2 \rangle\rangle$  modulo terms of total degree  $\geq p$ , being truncations of  $\exp \log(1 + t) = 1 + t$ ,  $\log \exp t = t$ ,  $\log((1 + t_1)(1 + t_2)) = \log(1 + t_1) + \log(1 + t_2)$  for commuting  $t_1, t_2$ , and  $\exp t_1 \exp t_2 = \exp H(t_1, t_2)$ . Both sides have coefficients in  $\Lambda$ , for (c) by fact (B) below. So they hold in  $\Lambda\langle t_1, t_2 \rangle / (t_1, t_2)^p$ , and in its commutative quotient for (b). Specialising  $t_i \mapsto z_i$  gives the claim.  $\square$

We call an element or an operator *integral* if it is defined over  $\Lambda$ . An identity between integral expressions that holds after extending scalars to  $\mathbb{Q}$  holds after reduction modulo  $p$ .

**2.2. Standard facts.** We use the following results, cited at the level of chapters.

- (J) *Jennings* [Jen]. Let  $X$  be a finite  $p$ -group and let  $M_n(X) = X \cap (1 + I_X^n)$ . Then  $M_n(X) = \prod_{i p^j \geq n} \gamma_i(X)^{p^j}$ . Hence every epimorphism  $X \rightarrow Y$  maps  $M_n(X)$  onto  $M_n(Y)$ . Moreover, if  $p^{d_n} = |M_n(X)/M_{n+1}(X)|$ , then

$$(1) \quad \sum_{n \geq 0} \dim(I_X^n / I_X^{n+1}) t^n = \prod_{n \geq 1} (1 + t^n + \dots + t^{(p-1)n})^{d_n}.$$

The coefficient of  $t^n$  is  $d_n$  plus a polynomial in  $d_1, \dots, d_{n-1}$ . So the dimensions of  $I_X^n / I_X^{n+1}$  for  $n \leq N$  determine  $d_1, \dots, d_N$ , and conversely.

- (R) *Restricted PBW and primitives* [Jac, Ch. V], [MM, Sec. 6]. Let  $\mathfrak{L}^{\text{res}} \subset \mathbb{F}_p\langle Y_1, \dots, Y_d \rangle$  be the restricted Lie subalgebra generated by  $Y_1, \dots, Y_d$ ; it is graded. Then  $\mathbb{F}_p\langle Y \rangle = U_{\text{res}}(\mathfrak{L}^{\text{res}})$ , and the following hold.
  - Let  $U_{\leq j}$  be the span of all products of at most  $j$  elements of  $\mathfrak{L}^{\text{res}}$ . Then  $\text{gr } U = \text{Sym}(\mathfrak{L}^{\text{res}})/(z^p : z \in \mathfrak{L}^{\text{res}})$ .
  - For the coproduct in which the  $Y_a$  are primitive, the primitive elements of  $\mathbb{F}_p\langle Y \rangle$  are exactly the elements of  $\mathfrak{L}^{\text{res}}$ .
  - With  $\ell_n = \dim \mathfrak{L}_n^{\text{res}}$ , the Hilbert series of  $\mathbb{F}_p\langle Y \rangle$  is  $\prod_{n \geq 1} (1 + t^n + \dots + t^{(p-1)n})^{\ell_n}$ .
- (Z) *PBW over  $\Lambda$*  [Bou, Ch. I–II]. The free Lie algebra  $\text{Lie}_\Lambda(t_1, \dots, t_r)$  is a free  $\Lambda$ -module, its universal enveloping algebra is  $\Lambda\langle t_1, \dots, t_r \rangle$ , and the PBW theorem holds for it.
- (B) *Dynkin's formula* [Dyn]. For  $n < p$ ,  $H_n$  is a Lie polynomial with coefficients in  $\Lambda$ .
- (L) *Lazard correspondence* [Laz], [Khu, Ch. 9–10]. Let  $\mathfrak{l}$  be a finite Lie ring of  $p$ -power order and nilpotency class  $< p$ . Then  $H_{<p}$  defines a group structure on  $\mathfrak{l}$ , and the subgroups of this group are exactly the Lie subrings of  $\mathfrak{l}$ .

**2.3. Standing hypothesis.** In Sections 4–7 we fix an integer  $N$  with

$$(2) \quad k \leq N \leq p + k - 2.$$

The upper bound is equivalent to each of  $p + k - 1 > N$ ,  $N - k + 1 \leq p - 1$  and  $N - k + 2 \leq p$ . Since  $k \geq 2$ , it implies the weaker inequality  $pk > N$ .

## 3. THE TRUNCATED FREE ALGEBRA

In this section  $N \geq 1$  is arbitrary. Let  $A = \mathbb{F}_p\langle X_1, \dots, X_d \rangle$ , let  $\mathfrak{m}$  be the ideal generated by  $X_1, \dots, X_d$ , and put  $T = A/\mathfrak{m}^{N+1}$ . Let  $A_\Lambda = \Lambda\langle X \rangle / \mathfrak{m}_\Lambda^{N+1}$ , where  $\mathfrak{m}_\Lambda$  is the augmentation ideal of  $\Lambda\langle X \rangle$ , and let  $A_\mathbb{Q} = A_\Lambda \otimes \mathbb{Q}$ . Then  $A_\Lambda / pA_\Lambda = T$ .

**Lemma 3.1.** (a) *The assignment  $x_a \mapsto 1 + X_a$  induces an isomorphism of algebras from  $\mathbb{F}_p F / I_F^{N+1}$  onto  $T$ .*

- (b) The subgroups  $M_n(F)$  are fully invariant in  $F$ . The group  $P = F/M_{N+1}(F)$  embeds in  $T^\times$  and is a finite  $p$ -group. Moreover  $T = \mathbb{F}_p P / I_P^{N+1}$ , and  $M_n(P) = P \cap (1 + \mathfrak{m}^n)$  for  $n \leq N+1$ ; in particular  $M_{N+1}(P) = 1$ .
- (c) Put  $Y_a = \ell(X_a)$ . Then  $A_\Lambda$  is the quotient of the free algebra  $\Lambda\langle Y_1, \dots, Y_d \rangle$  by the span of the  $Y$ -words of length  $> N$ . Similarly  $T$  is the quotient of  $\mathbb{F}_p\langle Y_1, \dots, Y_d \rangle$  by the span of the  $Y$ -words of length  $> N$ .

*Proof.* (a) The algebra map  $\mathbb{F}_p F \rightarrow T$  with  $x_a \mapsto 1 + X_a$  is surjective and kills  $I_F^{N+1}$ . Its inverse is induced by  $X_a \mapsto x_a - 1$ . This is defined because  $A$  is free and  $x_a - 1 \in I_F$ , so  $\mathfrak{m}^{N+1}$  is mapped into  $I_F^{N+1}$ . The two maps are inverse to each other on generators.

(b) Endomorphisms of  $F$  preserve  $I_F$ , hence they preserve  $M_n(F)$ . By (a) and the definition of  $M_{N+1}(F)$ , the map  $F \rightarrow T^\times$  has kernel  $M_{N+1}(F)$ . The group  $T^\times$  is finite. For  $z \in \mathfrak{m}$  and  $p^r > N$  we have  $(1+z)^{p^r} = 1 + z^{p^r} = 1$ , so  $P$  is a finite  $p$ -group. The kernel of  $\mathbb{F}_p F \rightarrow \mathbb{F}_p P$  is generated by the elements  $g - 1$  with  $g \in M_{N+1}(F)$ , and these lie in  $I_F^{N+1}$ . So  $\mathbb{F}_p P / I_P^{N+1} = \mathbb{F}_p F / I_F^{N+1} = T$ . Finally, the surjection  $\mathbb{F}_p P \rightarrow T$  has kernel  $I_P^{N+1}$  and maps  $I_P^n$  onto  $\mathfrak{m}^n$ , which gives the description of  $M_n(P)$ .

(c) The endomorphism of  $A_\Lambda$  with  $X_a \mapsto \ell(X_a)$  is the identity modulo  $\mathfrak{m}_\Lambda^2$ . So it is the identity plus a nilpotent map, hence invertible.  $\square$

We grade  $T$  by  $Y$ -length:  $T = \bigoplus_{n=0}^N T_n$ , where  $T_n$  is spanned by the  $Y$ -words of length  $n$ , and  $\mathfrak{m}^n = \bigoplus_{i \geq n} T_i$ . We do the same for  $A_\Lambda$  and  $A_\mathbb{Q}$ , and put  $V = T_1$ .

Note that  $\bar{X}_a \mapsto \ell(X_a)$  is not an automorphism of the untruncated algebra  $A$ . Its inverse on  $A_\Lambda$  differs from  $Y_a \mapsto \varepsilon(Y_a) - 1$  as soon as  $N \geq p$ .

**Lemma 3.2.** *Let  $\psi, \psi'$  be unital algebra endomorphisms of  $T$ , or of  $A_\mathbb{Q}$ , with  $\psi(Y_a), \psi'(Y_a) \in \mathfrak{m}$  and  $\psi(Y_a) - \psi'(Y_a) \in \mathfrak{m}^q$  for all  $a$ . Then  $(\psi - \psi')(\mathfrak{m}^r) \subseteq \mathfrak{m}^{q+r-1}$  for all  $r \geq 1$ . Likewise, a derivation  $\delta$  with  $\delta(Y_a) \in \mathfrak{m}^q$  for all  $a$  satisfies  $\delta(\mathfrak{m}^r) \subseteq \mathfrak{m}^{q+r-1}$ .*

*In particular, if  $q \geq N - k + 2$  then  $\psi = \psi'$  and  $\delta = 0$  on  $\mathfrak{m}^k$ . Also, if  $u$  is an algebra automorphism of  $T$  inducing the identity on  $\mathfrak{m}/\mathfrak{m}^2$ , then  $(u - 1)(\mathfrak{m}^n) \subseteq \mathfrak{m}^{n+1}$  for all  $n$ , and  $u$  fixes  $T_N$  pointwise.*

*Proof.* For a  $Y$ -word  $y_1 \cdots y_s$  with  $s \geq r$  we have

$$(\psi - \psi')(y_1 \cdots y_s) = \sum_i \psi(y_1) \cdots \psi(y_{i-1}) (\psi - \psi')(y_i) \psi'(y_{i+1}) \cdots \psi'(y_s) \in \mathfrak{m}^{q+s-1}.$$

The argument for  $\delta$  is the same. The last statement is the case  $\psi = u$ ,  $\psi' = \text{id}$ ,  $q = 2$ .  $\square$

**Notation.** Let  $\mathfrak{L} \subseteq T$  be the image of  $\bigoplus_{n \leq N} \mathfrak{L}_n^{\text{res}}$  (the restricted Lie subalgebra of  $T$  generated by the  $Y_a$ ), and put  $\mathfrak{L}_{\geq k} = \mathfrak{L} \cap \mathfrak{m}^k = \bigoplus_{n=k}^N \mathfrak{L}_n^{\text{res}}$ .

For a normal subgroup  $S$  of  $P$  with  $S \leq M_k(P)$  let:

- $\mathfrak{b}_S$  be the ideal of  $T$  generated by  $S - 1$ , so that  $T/\mathfrak{b}_S = \mathbb{F}_p[P/S]/I^{N+1}$ ;
- $M_S = \mathfrak{m}\mathfrak{b}_S + \mathfrak{b}_S\mathfrak{m}$ ;
- $\log s = \ell(s - 1)$  for  $s \in S$ , and  $\log S = \{\log s : s \in S\}$ .

#### 4. LOGARITHMS ON $M_k(P)$

We assume (2) from now on.

**Lemma 4.1.** *The map  $\log$  is a bijection from  $M_k(P)$  onto  $\mathfrak{L}_{\geq k}$ , with inverse  $\varepsilon$ . Moreover  $\varepsilon(z)\varepsilon(w) = \varepsilon(H_{<p}(z, w))$  for all  $z, w \in \mathfrak{m}^k$ .*

*Proof.* For  $s \in M_k(P)$  we have  $(s - 1)^j \in \mathfrak{m}^{jk} = 0$  for  $j \geq p$ , since  $pk > N$ . Similarly every product of  $p$  elements of  $\mathfrak{m}^k$  vanishes. So by Lemma 2.1 we have  $\varepsilon(\log s) = s$  and  $\log \varepsilon(z) = z$  for  $z \in \mathfrak{m}^k$ , and also  $\varepsilon(z)\varepsilon(w) = \varepsilon(H_{<p}(z, w))$ .

*Step 1: two coproducts.* Let  $\mathcal{F}_n = \sum_{i+j=n} \mathfrak{m}^i \otimes \mathfrak{m}^j \subseteq T \otimes T$ . This is the span of the tensors of  $Y$ -words of total length  $\geq n$ . Put  $T \hat{\otimes} T = (T \otimes T)/\mathcal{F}_{N+1}$ . By Lemma 3.1,  $T$  is

free on the  $X_a$  and also on the  $Y_a$ , modulo words of length  $> N$ . So there are algebra maps  $\Delta, \Delta_0: T \rightarrow T \hat{\otimes} T$  with

$$\Delta(X_a) = X_a \otimes 1 + 1 \otimes X_a + X_a \otimes X_a, \quad \Delta_0(Y_a) = Y_a \otimes 1 + 1 \otimes Y_a.$$

Since  $\Delta(x_a) = x_a \otimes x_a$ , we have  $\Delta(g) = g \otimes g$  for every  $g \in P$ . The same formulas define maps on  $A_\Lambda$  and  $A_\mathbb{Q}$ .

In  $A_\mathbb{Q}$  write  $\log(1 + X_a) = Y_a + c_a$  with  $c_a = \sum_{j=p}^N (-1)^{j+1} X_a^j / j \in \mathfrak{m}^p$ . The elements  $x_a \otimes 1$  and  $1 \otimes x_a$  commute and are unipotent, so  $\Delta(\log x_a) = \log(x_a \otimes x_a) = \log x_a \otimes 1 + 1 \otimes \log x_a$ . Hence

$$\Delta(Y_a) - \Delta_0(Y_a) = c_a \otimes 1 + 1 \otimes c_a - \Delta(c_a) \in \mathcal{F}_p.$$

The left-hand side is integral. The tensors of  $Y$ -words form a  $\Lambda$ -basis, and  $\mathcal{F}_p$  is spanned by a subset of it. So the left-hand side lies in the  $\Lambda$ -span of that subset, and after reduction we get  $\Delta(Y_a) - \Delta_0(Y_a) \in \mathcal{F}_p$  in  $T \hat{\otimes} T$ . Since  $\Delta(Y_b), \Delta_0(Y_b) \in \mathcal{F}_1$ , the telescoping argument of Lemma 3.2 gives  $(\Delta - \Delta_0)(\mathfrak{m}^k) \subseteq \mathcal{F}_{p+k-1}$ . This is 0 because  $p + k - 1 > N$ .

*Step 2: log s is primitive.* Let  $s \in M_k(P)$ , and put  $\alpha = (s - 1) \otimes 1$  and  $\beta = 1 \otimes (s - 1)$ . These commute, lie in  $\mathcal{F}_k$ , and satisfy  $\alpha^i \beta^j = 0$  for  $i + j \geq p$ . By Step 1,  $\Delta_0(s) = \Delta(s) = (1 + \alpha)(1 + \beta)$ . So by Lemma 2.1(b),

$$\Delta_0(\log s) = \ell(\alpha + \beta + \alpha\beta) = \ell(\alpha) + \ell(\beta) = \log s \otimes 1 + 1 \otimes \log s.$$

The map  $\Delta_0$  is graded, and in degrees  $\leq N$  the space  $T \hat{\otimes} T$  coincides with  $\mathbb{F}_p\langle Y \rangle \otimes \mathbb{F}_p\langle Y \rangle$ . Thus every homogeneous component of  $\log s$  is primitive in  $\mathbb{F}_p\langle Y \rangle$ , and so lies in  $\mathfrak{L}^{\text{res}}$  by (R). As  $\log s \in \mathfrak{m}^k$ , we get  $\log s \in \mathfrak{L}_{\geq k}$ .

*Step 3: bijectivity.* The map  $\log$  is injective on  $M_k(P)$ , since  $\varepsilon \circ \log = \text{id}$ . In degrees  $\leq N$  the graded algebra of  $T = \mathbb{F}_p P / I_P^{N+1}$  is  $\mathbb{F}_p\langle Y \rangle$ . By (1) for  $P$  and the Hilbert series in (R), the numbers  $d_n(P)$  and  $\dim \mathfrak{L}_n^{\text{res}}$  are obtained from the same Hilbert series by the same triangular recursion. Hence they agree for  $n \leq N$ . Since  $M_{N+1}(P) = 1$ , we get  $|M_k(P)| = p^{\sum_{n=k}^N d_n(P)} = |\mathfrak{L}_{\geq k}|$ , so  $\log$  is onto.  $\square$

**Lemma 4.2.** *Let  $S$  be a normal subgroup of  $P$  with  $S \leq M_k(P)$ . Then  $\log S$  is an  $\mathbb{F}_p$ -subspace of  $\mathfrak{L}_{\geq k}$ , and  $[\mathfrak{L}, \log S] \subseteq \log S$ .*

*Proof. Subspace.* With the commutator bracket of  $T$ ,  $\mathfrak{L}_{\geq k}$  is a Lie ring of class  $< p$ , since a bracket of  $p$  of its elements lies in  $\mathfrak{m}^{pk} = 0$ . On such a Lie ring  $H_{<p}$  is exactly Lazard's group law. By Lemma 4.1,  $\varepsilon$  is an isomorphism from this group onto  $M_k(P)$ . By (L), the preimage  $\log S$  of the subgroup  $S$  is a Lie subring, in particular an additive subgroup, and hence an  $\mathbb{F}_p$ -subspace.

*Ideal.* For  $b \leq d$  let  $c_b(y) = x_b y x_b^{-1}$ . This is an algebra automorphism of  $T$  with  $c_b(\log s) = \log(x_b s x_b^{-1})$ , so  $c_b(\log S) = \log S$ .

In  $A_\mathbb{Q}$  we have  $c_b = \exp(\text{ad } \log x_b)$ , and  $\log x_b = Y_b + c'_b$ , where  $c'_b \in \mathfrak{m}^p$  is a polynomial in  $X_b$ . Thus  $\text{ad } Y_b$  and  $\text{ad } c'_b$  commute, and  $\text{ad } c'_b$  maps  $\mathfrak{m}^k$  into  $\mathfrak{m}^{p+k} = 0$ . Hence  $c_b = \exp(\text{ad } Y_b)$  on  $\mathfrak{m}^k$ .

Both  $(\text{ad } Y_b)^j$  and  $(c_b - 1)^j$  vanish on  $\mathfrak{m}^k$  for  $j > N - k$ . So on  $\mathfrak{m}^k$

$$\text{ad } Y_b = \sum_{j=1}^{N-k} \frac{(-1)^{j+1}}{j} (c_b - 1)^j,$$

an integral identity because  $N - k \leq p - 2$ . Therefore  $\text{ad } Y_b$  preserves the subspace  $\log S$ .

The set of  $h \in \mathfrak{L}$  with  $[h, \log S] \subseteq \log S$  is a subspace, closed under brackets by the Jacobi identity and under  $h \mapsto h^p$  because  $\text{ad}(h^p) = (\text{ad } h)^p$ . It contains every  $Y_b$ , so it is all of  $\mathfrak{L}$ .  $\square$

## 5. THE LIE PART

For  $0 \leq j \leq p-1$  let  $\beta_j: \text{Sym}^j(\mathfrak{L}^{\text{res}}) \rightarrow \mathbb{F}_p\langle Y \rangle$  be symmetrisation,

$$\beta_j(z_1 \cdots z_j) = \frac{1}{j!} \sum_{\pi \in S_j} z_{\pi(1)} \cdots z_{\pi(j)}.$$

In particular  $\beta_0(\text{Sym}^0) = \mathbb{F}_p \cdot 1$  and  $\beta_1(\text{Sym}^1) = \mathfrak{L}^{\text{res}}$ .

**Lemma 5.1.**  $U_{\leq p-1} = \bigoplus_{j=0}^{p-1} \beta_j(\text{Sym}^j \mathfrak{L}^{\text{res}})$ , and all the summands are graded subspaces of  $\mathbb{F}_p\langle Y \rangle$ .

*Proof.* By (R),  $U_{\leq j}/U_{\leq j-1} \cong \text{Sym}^j(\mathfrak{L}^{\text{res}})$  for  $j \leq p-1$ , because the ideal  $(z^p)$  lives in degrees  $\geq p$  of  $\text{Sym}(\mathfrak{L}^{\text{res}})$ . Since  $\beta_j(z_1 \cdots z_j) \equiv z_1 \cdots z_j$  modulo  $U_{\leq j-1}$ , the map  $\beta_j$  lifts this isomorphism, so  $U_{\leq j} = U_{\leq j-1} \oplus \beta_j(\text{Sym}^j \mathfrak{L}^{\text{res}})$ . Gradedness is clear, since  $\mathfrak{L}^{\text{res}}$  is graded.  $\square$

**The projection  $f$ .** Fix a graded complement  $C$  of  $U_{\leq p-1}$  in  $\mathbb{F}_p\langle Y \rangle$ , and put

$$\widehat{B} = \bigoplus_{j=2}^{p-1} \beta_j(\text{Sym}^j \mathfrak{L}^{\text{res}}).$$

Let  $f$  be the projection of  $\mathbb{F}_p\langle Y \rangle$  onto  $\mathfrak{L}^{\text{res}}$  along  $\mathbb{F}_p \cdot 1 \oplus \widehat{B} \oplus C$ . Since  $f$  is graded, it induces a projection of  $T$  onto  $\mathfrak{L}$ , again denoted  $f$ . Let  $B$  be the image of  $\widehat{B}$  in  $T$ .

**Lemma 5.2.** (a)  $f$  fixes  $\mathfrak{L}$  and kills  $B$ .

(b) For  $1 \leq n \leq p-1$ ,  $T_n$  is contained in the image of  $\bigoplus_{j=1}^n \beta_j(\text{Sym}^j \mathfrak{L}^{\text{res}})$ .

(c)  $[\mathfrak{L}, B] \subseteq B$ , and likewise  $[\mathfrak{L}^{\text{res}}, \widehat{B}] \subseteq \widehat{B}$ .

(d) For  $2 \leq r \leq p-1$  there is a multilinear Lie polynomial  $e_r(t_1, \dots, t_r)$  with coefficients in  $\Lambda$  such that  $f(z_1 \cdots z_r) = e_r(z_1, \dots, z_r)$  for all  $z_1, \dots, z_r \in \mathfrak{L}$ . In particular, if  $\mathfrak{a}$  is a Lie ideal of  $\mathfrak{L}$  and some  $z_i \in \mathfrak{a}$ , then  $f(z_1 \cdots z_r) \in [\mathfrak{L}, \mathfrak{a}]$ .

*Proof.* (a) is the definition of  $f$ .

(b) A word of length  $n$  is a product of  $n$  elements of  $V \subseteq \mathfrak{L}^{\text{res}}$ , so it lies in  $U_{\leq n} = \bigoplus_{j \leq n} \beta_j(\text{Sym}^j \mathfrak{L}^{\text{res}})$ . Its component in  $\beta_0 = \mathbb{F}_p \cdot 1$  vanishes for degree reasons.

(c)  $\text{ad } w$  is a derivation, so  $\text{ad } w \beta_j(z_1 \cdots z_j) = \sum_i \beta_j(z_1 \cdots [w, z_i] \cdots z_j)$ .

(d) By (Z), and since  $r! \in \Lambda^\times$ , the filtration of  $\Lambda\langle t_1, \dots, t_r \rangle$  by the number of factors from  $\text{Lie}_\Lambda(t)$  satisfies

$$U_{\leq r} = \bigoplus_{j \leq r} \beta_j(\text{Sym}^j \text{Lie}_\Lambda(t)),$$

compatibly with the  $\mathbb{Z}^r$ -grading. Taking components of multidegree  $(1, \dots, 1)$ , we can write  $t_1 \cdots t_r = \sum_{j=1}^r E_j$  with  $E_j \in \beta_j(\text{Sym}^j \text{Lie}_\Lambda(t))$  multilinear. Put  $e_r = E_1$ .

Lift each  $z_i$  to an element of  $\bigoplus_{n \leq N} \mathfrak{L}_n^{\text{res}}$ . The algebra map  $t_i \mapsto z_i$  sends  $\text{Lie}_\Lambda(t)$  into  $\mathfrak{L}^{\text{res}}$  and  $\beta_j(\text{Sym}^j \text{Lie}_\Lambda(t))$  into  $\beta_j(\text{Sym}^j \mathfrak{L}^{\text{res}})$ . Hence  $z_1 \cdots z_r - e_r(z) \in \widehat{B}$ , and  $f(z_1 \cdots z_r) = e_r(z)$ .

A multilinear Lie polynomial in  $r \geq 2$  variables is a  $\Lambda$ -linear combination of bracket monomials containing each variable exactly once. Such a monomial with one entry in a Lie ideal  $\mathfrak{a}$  lies in  $[\mathfrak{L}, \mathfrak{a}]$ . This is clear by induction on the bracket length, since  $[\mathfrak{L}, \mathfrak{a}] \subseteq \mathfrak{a}$ .  $\square$

**Lemma 5.3.** Let  $S$  be a normal subgroup of  $P$  with  $S \leq M_k(P)$ , and put  $\mathfrak{a} = \log S$ . Then:

(a)  $\mathfrak{b}_S = \mathfrak{a} + M_S$ ;

(b)  $f(M_S) \subseteq [\mathfrak{L}, \mathfrak{a}] \subseteq \mathfrak{a} \cap M_S$ ;

(c)  $f(s-1) = \log s$  for  $s \in S$ ,  $f(\mathfrak{b}_S) = \mathfrak{a}$ , and  $(1-f)\mathfrak{b}_S \subseteq M_S$ .

*Proof.* (a) Let  $s \in S$  and  $z = \log s$ . Then  $z \in \mathfrak{b}_S$ , since  $z$  is a polynomial in  $s-1$  without constant term. By Lemma 2.1,  $s-1 = \varepsilon(z) - 1 = z + \sum_{j=2}^{p-1} z^j/j!$ , and  $z^j \in \mathfrak{b}_S \mathfrak{m} \subseteq M_S$  for

$j \geq 2$ . For  $a, b \in T$  write  $a = \alpha + a'$  and  $b = \beta + b'$  with  $\alpha, \beta \in \mathbb{F}_p$  and  $a', b' \in \mathfrak{m}$ . Then  $a(s-1)b \equiv \alpha\beta(s-1)$  modulo  $M_S$ . Hence

$$\mathfrak{b}_S = \text{span}\{s-1 : s \in S\} + M_S = \mathfrak{a} + M_S.$$

(b) Substitute (a) into  $M_S = \mathfrak{m}\mathfrak{b}_S + \mathfrak{b}_S\mathfrak{m}$  repeatedly. Since  $M_S \subseteq \mathfrak{m}^{k+1}$  and  $\mathfrak{m}^{N+1} = 0$ , this shows that  $M_S$  is contained in the span of the products  $z_1 \cdots z_r$  with  $r \geq 2$ ,  $z_i \in V \cup \mathfrak{a}$ , and at least one  $z_i \in \mathfrak{a}$ . Such a product lies in  $\mathfrak{m}^{k+r-1}$ , so it vanishes unless  $r \leq N-k+1 \leq p-1$ . By Lemma 4.2,  $\mathfrak{a}$  is a Lie ideal of  $\mathfrak{L}$ , so Lemma 5.2(d) gives  $f(M_S) \subseteq [\mathfrak{L}, \mathfrak{a}]$ . Also  $[\mathfrak{L}, \mathfrak{a}] \subseteq \mathfrak{a}$  by Lemma 4.2, and  $[y, z] \in \mathfrak{m}\mathfrak{b}_S + \mathfrak{b}_S\mathfrak{m}$  for  $y \in \mathfrak{L} \subseteq \mathfrak{m}$  and  $z \in \mathfrak{a} \subseteq \mathfrak{b}_S$ .

(c) For  $2 \leq j \leq p-1$  we have  $z^j/j! = \beta_j(z^j)/j! \in B$ . So  $f(s-1) = f(z) = z$  by Lemma 5.2(a). Then  $f(\mathfrak{b}_S) = \mathfrak{a} + f(M_S) = \mathfrak{a}$  by (a) and (b). Finally  $(1-f)\mathfrak{b}_S = (1-f)M_S \subseteq M_S + f(M_S) \subseteq M_S$ .  $\square$

## 6. UNIPOTENT AUTOMORPHISMS

**Lemma 6.1.** *Let  $u$  be an algebra automorphism of  $T$  inducing the identity on  $\mathfrak{m}/\mathfrak{m}^2$ . Then there is a derivation  $D$  of  $T$  such that:*

- (a)  $D(Y_a) \in T_2 \oplus \cdots \oplus T_{N-k+1}$  for every  $a$ ;
- (b)  $D = \sum_{j=1}^{N-k} \frac{(-1)^{j+1}}{j} (u-1)^j$  on  $\mathfrak{m}^k$ ;
- (c)  $u = \sum_{j=0}^{N-k} D^j/j!$  on  $\mathfrak{m}^k$ ;
- (d)  $D$  preserves every  $u$ -stable subspace of  $\mathfrak{m}^k$ .

This lemma only uses  $N-k \leq p-1$ .

*Proof.* Lift each  $u(Y_a)$  to an element of  $Y_a + \mathfrak{m}_\Lambda^2$ , and let  $\tilde{u}$  be the endomorphism of  $A_\Lambda$  with these values on the  $Y_a$ . It is the identity plus a nilpotent map, hence an automorphism, and it reduces to  $u$ .

In  $A_\mathbb{Q}$  put  $\tilde{D} = \log \tilde{u} = \sum_{j \geq 1} (-1)^{j+1} (\tilde{u}-1)^j/j$ ; the sum is finite. Then  $\tilde{u} = \exp \tilde{D}$ . To see that  $\tilde{D}$  is a derivation, note that  $\tilde{u}^t = \sum_j \binom{t}{j} (\tilde{u}-1)^j$  is multiplicative for  $t \in \mathbb{N}$ , hence identically in  $t$ ; differentiating at  $t=0$  gives the Leibniz rule.

The component of  $(\tilde{u}-1)^j Y_a$  in degree  $n$  vanishes for  $n \leq j$ . So the components of  $\tilde{D}(Y_a)$  in degrees  $\leq N-k+1$  involve only  $j \leq N-k \leq p-1$ , and are integral. Let  $\tilde{D}'$  be the derivation of  $A_\mathbb{Q}$  sending  $Y_a$  to these components, and let  $D$  be its reduction modulo  $p$ . This gives (a).

The derivation  $\tilde{D} - \tilde{D}'$  sends each  $Y_a$  into  $\mathfrak{m}^{N-k+2}$ , so it vanishes on  $\mathfrak{m}^k$  by Lemma 3.2. Moreover  $(\tilde{u}-1)^j$  and  $\tilde{D}^j$  vanish on  $\mathfrak{m}^k$  for  $j > N-k$ . Hence on  $\mathfrak{m}^k$

$$\tilde{D}' = \sum_{j=1}^{N-k} \frac{(-1)^{j+1}}{j} (\tilde{u}-1)^j, \quad \tilde{u} = \sum_{j=0}^{N-k} \frac{(\tilde{D}')^j}{j!}.$$

Both identities are integral, and reducing them modulo  $p$  gives (b) and (c). Part (d) follows from (b).  $\square$

**Lemma 6.2.** *Let  $\rho_a \in \mathfrak{L} \cap (T_2 \oplus \cdots \oplus T_{p-1})$  for  $a = 1, \dots, d$ . Then there is  $\sigma \in \text{Aut}(P)$ , inducing the identity on  $P/\Phi(P)$ , such that  $\sigma(Y_a) \equiv Y_a + \rho_a$  modulo  $\mathfrak{m}^p$  for every  $a$ . Consequently, if  $\psi_\rho$  is the algebra endomorphism of  $T$  with  $Y_a \mapsto Y_a + \rho_a$ , then  $\sigma = \psi_\rho$  on  $\mathfrak{m}^k$ .*

*Proof.* If  $p=2$ , then  $\rho_a \in T_2 \oplus \cdots \oplus T_1 = 0$  and  $\sigma = \text{id}$  works. So let  $p \geq 3$ .

Put  $T^0 = A/\mathfrak{m}^p = \mathbb{F}_p F/I_F^p$  and  $P^0 = F/M_p(F)$ . Lemmas 3.1 and 4.1 apply with  $(k, N)$  replaced by  $(2, p-1)$ , which satisfies (2) because  $p \geq 3$ . So  $\varepsilon$  is a bijection from  $\mathfrak{L}_{\geq 2}^0$  onto  $M_2(P^0)$ , where  $\mathfrak{L}_{\geq 2}^0$  is the space of Lie elements of  $T^0$  of degrees  $2, \dots, p-1$ . We regard  $\rho_a$  as an element of  $\bar{T}^0$  via its homogeneous components, all of which have degree  $< p$ .

In  $T^0$  every product of  $p$  elements of  $\mathfrak{m}$  vanishes. So by Lemma 2.1,  $\varepsilon$  and  $\ell$  are inverse bijections between  $\mathfrak{m}$  and  $1 + \mathfrak{m}$ ,  $\varepsilon(Y_a) = x_a$ , and  $\varepsilon(z)\varepsilon(w) = \varepsilon(H_{<p}(z, w))$  for all  $z, w \in \mathfrak{m}$ .

Put  $g_a = H_{<p}(-Y_a, Y_a + \rho_a)$ . This is a Lie element with no component in degree 1, so  $g_a \in \mathfrak{L}_{\geq 2}^0$  and  $\varepsilon(g_a) \in M_2(P^0)$ . Moreover

$$x_a \varepsilon(g_a) = \varepsilon(Y_a) \varepsilon(-Y_a) \varepsilon(Y_a + \rho_a) = \varepsilon(Y_a + \rho_a).$$

Choose  $\hat{s}_a \in M_2(F)$  mapping to  $\varepsilon(g_a)$ . Since  $M_{N+1}(F)$  is fully invariant, the endomorphism  $x_a \mapsto x_a \hat{s}_a$  of  $F$  induces an endomorphism  $\sigma$  of  $P$ . It is surjective modulo  $\Phi(P) = M_2(P)$ , hence an automorphism, and it induces the identity on  $P/\Phi(P)$ .

The endomorphism  $x_a \mapsto x_a \hat{s}_a$  of  $F$  also acts compatibly on every quotient  $A/\mathfrak{m}^n = \mathbb{F}_p F/I_F^n$ . So the congruence  $\sigma(Y_a) \equiv Y_a + \rho_a$  modulo  $\mathfrak{m}^p$  can be checked in  $T^0$ , whether  $N+1$  is larger or smaller than  $p$ . In  $T^0$  we have  $\sigma(x_a) = \varepsilon(Y_a + \rho_a)$ , hence  $\sigma(Y_a) = \ell(\sigma(x_a) - 1) = Y_a + \rho_a$ .

The last statement follows from Lemma 3.2 with  $q = p$ , since  $p \geq N - k + 2$ .  $\square$

## 7. THE UNIPOTENT STEP

**Proposition 7.1.** *Let  $S$  and  $S''$  be normal subgroups of  $P$  contained in  $M_k(P)$ . Let  $u$  be an algebra automorphism of  $T$  inducing the identity on  $\mathfrak{m}/\mathfrak{m}^2$ , with*

$$u(\mathfrak{b}_S) = \mathfrak{b}_{S''} \quad \text{and} \quad \mathfrak{b}_S + T_N = \mathfrak{b}_{S''} + T_N.$$

*Then there is  $\sigma \in \text{Aut}(P)$ , inducing the identity on  $P/\Phi(P)$ , with  $\sigma(S) = S''$ .*

*Proof.* Put  $\mathfrak{c} = \mathfrak{b}_S + T_N \subseteq \mathfrak{m}^k$  and  $M = \mathfrak{m}\mathfrak{c} + \mathfrak{c}\mathfrak{m}$ . Since  $\mathfrak{m}T_N = T_N\mathfrak{m} = 0$ , we have  $M = M_S = M_{S''}$ .

- By Lemma 3.2,  $u$  fixes  $T_N$  pointwise. It maps  $\mathfrak{b}_S$  onto  $\mathfrak{b}_{S''}$ , so it preserves  $\mathfrak{c}$ , and hence  $M$ .
- Let  $D$  be as in Lemma 6.1. Then  $D$  preserves  $\mathfrak{c}$  and  $M$ , and  $D(T_N) = 0$  because  $D$  raises degree.
- By Lemma 6.1(a) and Lemma 5.2(b),  $D(Y_a)$  lies in the image of  $\bigoplus_{j=1}^{p-1} \beta_j(\text{Sym}^j \mathfrak{L}^{\text{res}})$ . Put  $\lambda_a = f(D(Y_a)) \in \mathfrak{L}$  and  $\nu_a = D(Y_a) - \lambda_a \in B$ .
- Let  $D_L$  be the derivation of  $T$  with  $D_L(Y_a) = \lambda_a$ , and put  $E = D - D_L$ .

*Claim (1).*  $D_L(\mathfrak{L}) \subseteq \mathfrak{L}$  and  $E(\mathfrak{L}) \subseteq B$ . Consequently  $f(Dy) = D_L(y)$  for every  $y \in \mathfrak{L}$ .

Lift  $D_L$  and  $E$  to derivations  $\hat{D}_L$  and  $\hat{E}$  of  $\mathbb{F}_p\langle Y \rangle$  with the same values on the  $Y_a$ . These raise degree, so they induce  $D_L$  and  $E$  on  $T$ . Let  $\mathfrak{K}$  be the set of  $z \in \mathfrak{L}^{\text{res}}$  with  $\hat{D}_L z \in \mathfrak{L}^{\text{res}}$  and  $\hat{E}z \in \hat{B}$ . Then:

- $\mathfrak{K}$  contains every  $Y_a$ .
- $\mathfrak{K}$  is closed under brackets, by the Leibniz rule and Lemma 5.2(c).
- $\mathfrak{K}$  is closed under  $z \mapsto z^p$ . Indeed, every derivation  $\delta$  of an  $\mathbb{F}_p$ -algebra satisfies  $\delta(z^p) = \sum_{i=0}^{p-1} z^i \delta(z) z^{p-1-i} = (\text{ad } z)^{p-1}(\delta z)$ , since  $(L_z - R_z)^{p-1} = \sum_i L_z^i R_z^{p-1-i}$  in characteristic  $p$ .

So  $\mathfrak{K} = \mathfrak{L}^{\text{res}}$ . Projecting to  $T$  gives the two inclusions, and the last statement follows from Lemma 5.2(a).

*Claim (2).* For  $x \in \mathfrak{c}$  and  $j \geq 0$  we have  $f(D^j x) \equiv D_L^j f(x)$  modulo  $f(M)$ .

For  $z \in \mathfrak{c}$  we have  $z - f(z) \in M + T_N$ , by Lemma 5.3(c) and  $f(T_N) \subseteq T_N$ . Also  $f(\mathfrak{c}) \subseteq \mathfrak{c} \cap \mathfrak{L}$ , since  $f(\mathfrak{b}_S) = \log S \subseteq \mathfrak{b}_S$ . Hence

$$f(Dz) - f(Df(z)) \in f(D(M) + D(T_N)) \subseteq f(M) \quad (z \in \mathfrak{c}).$$

We induct on  $j$ . Suppose  $f(D^j x) = D_L^j f(x) + \mu$  with  $\mu \in f(M)$ . Apply the display to  $z = D^j x \in \mathfrak{c}$ :

$$f(D^{j+1} x) \equiv f(D D_L^j f(x)) + f(D\mu) = D_L^{j+1} f(x) + f(D\mu).$$

The equality is Claim 1, since  $D_L^j f(x) \in \mathfrak{L}$ . Finally  $f(D\mu) \in f(M)$ , because  $f(M) \subseteq M$  by Lemma 5.3(b) and  $D(M) \subseteq M$ .

*Claim (3).* There is  $\sigma \in \text{Aut}(P)$ , inducing the identity on  $P/\Phi(P)$ , with  $f(ux) \equiv \sigma(f(x))$  modulo  $f(M)$  for every  $x \in \mathfrak{c}$ .

Let  $\rho_a$  be the sum of the components of degree  $\leq N - k + 1$  of  $\sum_{j=1}^{N-k} D_L^j(Y_a)/j!$ . By Claim 1 and since  $N - k + 1 \leq p - 1$ , we have  $\rho_a \in \mathfrak{L} \cap (T_2 \oplus \cdots \oplus T_{p-1})$ . Let  $\sigma$  be given by Lemma 6.2. We show that

$$\sum_{j=0}^{N-k} \frac{D_L^j}{j!} = \psi_\rho = \sigma \quad \text{on } \mathfrak{m}^k.$$

- Lift  $D_L$  to a derivation  $\tilde{D}_L$  of  $A_\Lambda$  whose values on the  $Y_a$  lie in degrees  $2, \dots, N - k + 1$ . In  $A_\mathbb{Q}$ ,  $\exp \tilde{D}_L$  is an algebra automorphism, and on  $\mathfrak{m}^k$  it equals  $\sum_{j=0}^{N-k} \tilde{D}_L^j/j!$ .
- By Lemma 3.2, on  $\mathfrak{m}^k$  it also equals the endomorphism  $Y_a \mapsto Y_a + \tilde{\rho}_a$ , where  $\tilde{\rho}_a$  is the sum of the components of degree  $\leq N - k + 1$  of  $\sum_{j \geq 1} \tilde{D}_L^j(Y_a)/j!$ . Only  $j \leq N - k$  contributes to  $\tilde{\rho}_a$ .
- Both descriptions are integral. Reducing modulo  $p$  gives the first equality, and Lemma 6.2 gives the second.

By Lemma 6.1(c) and Claim 2, for  $x \in \mathfrak{c} \subseteq \mathfrak{m}^k$ ,

$$f(ux) = \sum_{j=0}^{N-k} \frac{f(D^j x)}{j!} \equiv \sum_{j=0}^{N-k} \frac{D_L^j f(x)}{j!} = \sigma(f(x)) \pmod{f(M)},$$

where the last equality uses  $f(x) \in \mathfrak{c} \subseteq \mathfrak{m}^k$ .

*Conclusion.* Let  $s \in S$ . Then  $s - 1 \in \mathfrak{c}$  and  $f(s - 1) = \log s$  by Lemma 5.3(c), so Claim 3 gives  $\sigma(\log s) \equiv f(u(s - 1))$  modulo  $f(M)$ .

- $u(s - 1) \in \mathfrak{b}_{S''}$ , so  $f(u(s - 1)) \in \log S''$  by Lemma 5.3(c) applied to  $S''$ .
- $f(M) = f(M_{S''}) \subseteq \log S''$  by Lemma 5.3(b) applied to  $S''$ .
- Hence  $\log \sigma(s) = \sigma(\log s) \in \log S''$ , and so  $\sigma(s) \in S''$  by Lemma 4.1.

Thus  $\sigma(S) \subseteq S''$ .

To see equality, note that  $u$  induces an isomorphism  $T/\mathfrak{b}_S \rightarrow T/\mathfrak{b}_{S''}$  preserving the  $\mathfrak{m}$ -adic filtrations. These algebras are  $\mathbb{F}_p[P/S]/I^{N+1}$  and  $\mathbb{F}_p[P/S'']/I^{N+1}$ , and  $M_{N+1}(P/S) = M_{N+1}(P/S'') = 1$  by (J). So (1) gives  $|P/S| = |P/S''|$ , and therefore  $\sigma(S) = S''$ .  $\square$

## 8. PROOF OF THEOREM A

**Proposition 8.1.** *Let  $k - 1 \leq N \leq p + k - 2$ , and let  $S$  and  $S'$  be normal subgroups of  $P = F/M_{N+1}(F)$  contained in  $M_k(P)$ . Let  $\theta$  be an algebra automorphism of  $T$  with  $\theta(\mathfrak{b}_S) = \mathfrak{b}_{S'}$ . Then there is  $\tau \in \text{Aut}(P)$  with  $\tau(S) = S'$  that induces on  $P/\Phi(P) \cong \mathfrak{m}/\mathfrak{m}^2$ ,  $g\Phi(P) \mapsto g - 1 + \mathfrak{m}^2$ , the same map as  $\theta$ .*

Here  $\theta$  preserves  $\mathfrak{m}$ , because  $\mathfrak{m}$  is the unique maximal ideal of  $T$ .

*Proof.* We induct on  $N$ .

*Base case*  $N = k - 1$ . Then  $M_k(P) = M_{N+1}(P) = 1$ , so  $S = S' = 1$ . Choose  $w_a \in F$  with  $w_a - 1 \equiv \theta(X_a)$  modulo  $\mathfrak{m}^2$ . Since  $M_{N+1}(F)$  is fully invariant,  $x_a \mapsto w_a$  induces an endomorphism of  $P$ . It is surjective modulo  $\Phi(P)$ , hence an automorphism, and it has the required linear part.

*Inductive step*,  $k \leq N \leq p + k - 2$ . Put  $\bar{T} = T/T_N = A/\mathfrak{m}^N$  and  $\bar{P} = F/M_N(F) = P/M_N(P)$ .

- The image of  $\mathfrak{b}_S$  in  $\bar{T}$  is  $\mathfrak{b}_{\bar{S}}$ , where  $\bar{S} = SM_N(P)/M_N(P) \leq M_k(\bar{P})$ . The automorphism  $\theta$  induces  $\bar{\theta}$  on  $\bar{T}$  with  $\bar{\theta}(\mathfrak{b}_{\bar{S}}) = \mathfrak{b}_{\bar{S}'}$ .
- By induction there is  $\bar{\tau}_0 \in \text{Aut}(\bar{P})$  with  $\bar{\tau}_0(\bar{S}) = \bar{S}'$  and the linear part of  $\theta$ .

- Lift  $\bar{\tau}_0$  to  $\tau_0 \in \text{Aut}(P)$  as in the base case, choosing  $w_a$  in the class  $\bar{\tau}_0(x_a M_N(F))$ . Then  $\tau_0(SM_N(P)) = S'M_N(P)$ , since both sides contain  $M_N(P)$ .
- Put  $u = \tau_0^{-1}\theta$  and  $S'' = \tau_0^{-1}(S')$ . Then  $S''$  is normal in  $P$  and contained in  $M_k(P)$ ,  $u$  induces the identity on  $\mathfrak{m}/\mathfrak{m}^2$ , and  $u(\mathfrak{b}_S) = \mathfrak{b}_{S''}$ .
- $S''M_N(P) = SM_N(P)$ . Hence  $\mathfrak{b}_S + T_N = \mathfrak{b}_{S''} + T_N$ , since both are the preimage of  $\mathfrak{b}_{\bar{S}}$  in  $T$ .

Proposition 7.1 gives  $\sigma \in \text{Aut}(P)$  with trivial linear part and  $\sigma(S) = S''$ . Then  $\tau = \tau_0\sigma$  has the required properties.  $\square$

*Proof of Theorem A.* Let  $N = p + k - 2$ , and let  $\varphi: \mathbb{F}_p G/I_G^{N+1} \rightarrow \mathbb{F}_p H/I_H^{N+1}$  be an isomorphism of algebras.

*Step 1: the hypothesis holds for  $H$ .* We have  $d(H) = \dim I_H/I_H^2 = \dim I_G/I_G^2 = d(G) = d$ . By (1), the dimensions of  $I^n/I^{n+1}$  for  $n < k$  give  $|H/M_k(H)| = |G/M_k(G)| = |F/M_k(F)|$ .

*Step 2: reduction to  $M_{N+1} = 1$ .* Replacing  $G$  by  $G/M_{N+1}(G)$  changes neither  $\mathbb{F}_p G/I_G^{N+1}$ , since  $M_{N+1}(G) - 1 \subseteq I_G^{N+1}$ , nor  $G/M_n(G)$  for  $n \leq N + 1$ , by (J). So we may assume  $M_{N+1}(G) = M_{N+1}(H) = 1$ .

*Step 3: presentations.* Sending  $x_1, \dots, x_d$  to a minimal generating set of  $G$  defines an epimorphism  $F \rightarrow G$ . It maps  $M_{N+1}(F)$  into  $M_{N+1}(G) = 1$ , so it factors through an epimorphism  $P \rightarrow G$ ; let  $S_G$  be its kernel.

- By (J),  $P/S_G M_k(P) \cong G/M_k(G)$ . This group has the order of  $F/M_k(F) = P/M_k(P)$ , so  $S_G \leq M_k(P)$ .
- By Step 1, the same holds for  $H$ .
- $\mathbb{F}_p G/I_G^{N+1} = T/\mathfrak{b}_{S_G}$ , and likewise for  $H$ .

*Step 4: lifting  $\varphi$*  (a truncated form of R hl's lifting [R 87, R 90]). Let  $\theta$  be the algebra endomorphism of  $T$  sending  $X_a$  to a preimage in  $\mathfrak{m}$  of  $\varphi(X_a + \mathfrak{b}_{S_G})$ . Then  $\theta(\mathfrak{b}_{S_G}) \subseteq \mathfrak{b}_{S_H}$ . Moreover  $\theta$  is surjective modulo  $\mathfrak{m}^2 + \mathfrak{b}_{S_H} = \mathfrak{m}^2$ , hence surjective, hence bijective. Comparing dimensions gives  $\theta(\mathfrak{b}_{S_G}) = \mathfrak{b}_{S_H}$ .

*Step 5: conclusion.* By Proposition 8.1, some  $\tau \in \text{Aut}(P)$  maps  $S_G$  onto  $S_H$  and has the linear part of  $\theta$ . So  $\tau$  induces an isomorphism  $G = P/S_G \rightarrow P/S_H = H$ , and its linear part corresponds to that of  $\varphi$ . Undoing Step 2,  $G/M_{p+k-1}(G) \cong H/M_{p+k-1}(H)$ .  $\square$

## 9. REMARKS

*Remark 9.1* (Where the window is used). The bound  $N \leq p + k - 2$  is used in the equivalent forms  $p + k - 1 > N$ ,  $N - k + 1 \leq p - 1$  and  $N - k + 2 \leq p$ , in these places:

- the two coproducts agree on  $\mathfrak{m}^k$  (Lemma 4.1);
- the tail  $c'_b$  of  $\log x_b$  acts trivially on  $\mathfrak{m}^k$  (Lemma 4.2);
- every product spanning  $M_S$  has fewer than  $p$  factors (Lemma 5.3);
- $D(Y_a)$  lives in degrees  $< p$ , where it has a Lie part (Lemma 5.2(b), Proposition 7.1);
- Lie substitutions need only be controlled modulo  $\mathfrak{m}^p$  (Lemma 6.2).

It is also used in the weaker form  $pk > N$ , for the Lazard correspondence on  $M_k(P)$  and for the exponential on  $\mathfrak{m}^k$  (Lemmas 4.1 and 4.2). Lemma 6.1 alone only needs  $N - k \leq p - 1$ .

At  $N = p + k - 1$  the bound fails by one. Then  $pk > N$  still holds, but every use in the list above breaks:

- a relator of degree  $k$  multiplied by  $p - 1$  generators is a product of  $p$  factors;
- $D(Y_a)$  can have a component of degree  $p$ , where there is no canonical Lie part;
- a Lie substitution must be realised modulo  $\mathfrak{m}^{p+1}$ .

We do not know whether the conclusion of Theorem A extends beyond the window.

*Remark 9.2* (Why only on  $\mathfrak{m}^k$ ). The restriction to  $\mathfrak{m}^k$  is essential. For example, let  $p = 3$ ,  $k = 3$  and  $N = 4$ , and let  $u$  be the automorphism of  $T$  with  $Y_1 \mapsto Y_1 + Y_2 Y_1$  and  $Y_2 \mapsto Y_2$ . Then  $(u-1)^3 Y_1 = Y_2^3 Y_1 \neq 0$ , so  $\log u$  requires division by 3 on  $T$ . On  $\mathfrak{m}^3$ , however,  $(u-1)^2 = 0$  and  $\log u = u - 1$ .

More generally,  $\log u$  is  $p$ -integral on all of  $T$  for every unipotent  $u$  exactly when  $N \leq p$ . If  $N \geq p + 1$ , the same automorphism gives a term  $\pm Y_2^p Y_1/p$  in  $\log u(Y_1)$ . Within the window this happens precisely when  $k \geq 3$ .

*Remark 9.3* (The case  $p = 2$ ). For  $p = 2$ , Theorem A adds a single layer, and it is Röhl's theorem [HS, Theorem 5.8]. It can also be seen directly.

Let  $L(G) = \bigoplus_n M_n(G)/M_{n+1}(G)$  be the restricted Lie algebra of the Jennings series. By Quillen [Qui],  $\text{gr } \mathbb{F}_p G \cong U_{\text{res}}(L(G))$ , and  $L(G)$  is generated by its degree-one part. If  $G$  is relatively free modulo  $M_k$ , then  $L(G)$  agrees with the free restricted Lie algebra in degrees  $< k$ . The group  $G/M_{k+1}(G)$  is then determined by the  $GL_d(\mathbb{F}_p)$ -orbit of the kernel of  $\mathfrak{L}_k^{\text{res}} \rightarrow L_k(G)$ , and  $\text{gr } \mathbb{F}_p G/I^{k+1}$  recovers this orbit.

In the proof above,  $p = 2$  forces  $N = k$  in the inductive step. Then  $D = 0$  and  $\sigma = \text{id}$ , so Proposition 7.1 says that  $S = S''$ .

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